LESSON 01 / 04

The theorem is a census of bracket choices

Expanding $(a+b)^n$ means picking one letter from each of n brackets. A pick-list is determined by which k brackets contributed b: $\binom{n}{k}$ lists, each giving $a^{n-k}b^k$. Hence, for non-negative integer n,

$$(a+b)^n = \sum_{k=0}^{n} \binom{n}{k} a^{n-k}b^k, \qquad T_{k+1} = \binom{n}{k} a^{n-k}b^k$$

No induction, nothing to memorise: the coefficient **is** a count (chapter 8). Symmetry $\binom{n}{k} = \binom{n}{n-k}$ (choose the a-brackets instead) and Pascal's rule (condition on the first bracket) are the same picture. Row n of Pascal's triangle is the census for $(a+b)^n$.

KEY RESULT

The general-term machine

For $(\alpha x^p + \beta x^{-q})^n$ the general term is $\binom{n}{k}\alpha^{n-k}\beta^k x^{p(n-k)-qk}$. "Coefficient of x^m" and "term independent of x" are one step: set the exponent $pn-(p+q)k$ equal to the target, solve for k; integer in $[0,n] \Rightarrow$ substitute; otherwise **the answer is 0** — "no such term" is a legitimate exam answer.

WORKED EXAMPLE

Worked — the machine at three gears

Coefficient of x^4 in $(2+x)^5$: $k=4$; $\binom{5}{4}2^1 = 10$ (full row $32+80x+80x^2+40x^3+10x^4+x^5$ ✓).

Term independent of x in $\left(x-\frac{2}{x^2}\right)^9$: exponent $9-3k=0$, $k=3$; $\binom{9}{3}(-2)^3 = -672$.

Coefficient of x^4 in $(3-2x)^7$: $k=4$; $\binom{7}{4}3^3(-2)^4 = 35\cdot 27\cdot 16 = 15120$ (even k, positive ✓).

KEY RESULT

Middle terms and equal coefficients

n even: one middle term $T_{n/2+1}$. n odd: two, at $k=\frac{n-1}{2}, \frac{n+1}{2}$. Equal coefficients: $\binom{n}{r} = \binom{n}{s}$ iff $r=s$ or $r+s=n$ — always report both branches: in $(1+x)^{18}$, coefficients of x^{2r} and x^{r+6} equal gives $2r=r+6$ or $2r+r+6=18$, so $r=6$ or $r=4$.

Consecutive-coefficient ratios: $\frac{\binom{n}{k+1}}{\binom{n}{k}} = \frac{n-k}{k+1}$ (three consecutive coefficients $36, 84, 126$ force row $n=9$; ratios $\frac{7}{3}, \frac{3}{2}$ ✓).

COMMON TRAP

Common traps — general term

- T_k vs T_{k+1}: the 5th term has $k=4$.
- Buried negatives: rewrite the bracket as $A+B$ with the minus inside B, then apply the formula robotically.
- Expanding term-by-term instead of writing T_{k+1}: the census view gives each term in one line.

ASSUMPTIONS & CONVENTIONS

Standing conventions for this chapter

Positive integral index n in Part I (official syllabus); general real index is Part II. $\binom{n}{k} = 0$ outside $0 \le k \le n$. Coefficients of an expansion are read after collecting like powers — "number of terms" questions must state in which variable(s) collection happens.

LESSON 02 / 04

Beyond two outcomes per bracket

KEY RESULT

Multinomial coefficients

In $(x_1 + \cdots + x_r)^n$, the coefficient of $x_1^{n_1} \cdots x_r^{n_r}$ $(\sum n_i = n)$ is $\frac{n!}{n_1! \cdots n_r!}$: line up the brackets, mark which gave each outcome. Number of raw products for $(a+b+c)^6$: $3^6 = 729$; collected terms (exponent tuples, stars and bars): $\binom{6+2}{2} = 28$. Coefficient of $a^2b^2c^2$ in $(a+b+c)^6$: $\frac{6!}{2!\,2!\,2!} = 90$.

Coefficient of x^3 in $(1+x+x^2)^4$ by census: patterns $\{x^2, x, 1, 1\}$ in $\frac{4!}{1!\,1!\,2!} = 12$ ways and $\{x, x, x, 1\}$ in $\frac{4!}{3!\,1!} = 4$ ways: total 16 (matches $(1+x+x^2)^4 = 1 + 4x + 10x^2 + 16x^3 + 19x^$ ✓).

COMMON TRAP

Number-of-terms trap

"Terms in $(1+x+x^2)^n$" is **not** $\binom{n+2}{2}$: collecting in the single variable x collapses to powers $x^0, \ldots, x^{2n}$ — exactly $2n+1$ terms. Always ask: collected in which variables?

KEY RESULT

Where coefficients peak

$\frac{\binom{n}{k+1}}{\binom{n}{k}} = \frac{n-k}{k+1}$ passes 1 at $k = \frac{n-1}{2}$. So row n: n even ⇒ single peak $\binom{n}{n/2}$; n odd ⇒ twin peaks $\binom{n}{(n-1)/2} = \binom{n}{(n+1)/2}$ (for $(1+x)^{13}$: 1716 at x^6, x^7; for $(1+x)^{21}$: 352716 at $k = 10, 11$). Coefficient of x^5 in $(1+x)^{12}$ is $\binom{12}{5} = 792$, and of x^7 also 792 by symmetry.

KEY RESULT

Substitution engine — coefficient sums without expanding

For $f(x) = (px+q)^n$ expanded: sum of coefficients $= f(1) = (p+q)^n$; constant term $= f(0) = q^n$; alternating sum $= f(-1)$ splits even/odd powers. Sum of coefficients of $(2x-1)^7$ is $(2-1)^7 = 1$; of $(x-2y)^9$ (set $x = y = 1$): $(-1)^9 = -1$ — negative answers are legal. Odd-power coefficients of $(1+x)^{12}$: $\frac{f(1)-f(-1)}{2} = 2^{11} = 2048$.

WORKED EXAMPLE

Worked — coefficient extraction across factors

Coefficient of x^8 in $(1+x)^{10}(1+\frac{1}{x})^{12}$: write the second factor as $x^{-12}(1+x)^{12}$; the product is $x^{-12}(1+x)^{22}$; need x^{20}: $\binom{22}{20} = 231$. Always clear negative powers into an x^{-m} prefactor first.

LESSON 03 / 04

One master polynomial, three engines

Everything runs on $f(x) = (1+x)^n = \sum_k \binom{n}{k} x^k$.

Engine 1 (substitute): $f(1) = 2^n$; $f(-1) = 0$ (even/odd halves 2^{n-1} each); $f(2) = 3^n$, i.e. $\sum \binom{n}{k} 2^k = 59049$ at $n = 10$. **Engine 2 (differentiate/integrate):** the operator $x\frac{d}{dx}$ stamps k onto each term:

$$\sum k\binom{n}{k} = n2^{n-1}, \qquad \sum k(k-1)\binom{n}{k} = n(n-1)2^{n-2}$$

$$\sum k^2\binom{n}{k} = n(n+1)2^{n-2}$$

($k^2 = k(k-1) + k$; at $n = 8$: $8 \cdot 9 \cdot 2^6 = 4608$.) Divide-by-$(k+1)$ is the integral's fingerprint: $\sum \frac{\binom{n}{k}}{k+1} = \int_0^1 (1+x)^n dx = \frac{2^{n+1}-1}{n+1}$, and $\sum \frac{(-1)^k}{k+1}\binom{n}{k} = \frac{1}{n+1}$.

INTUITION

Every identity is also a counting story

$\sum k\binom{n}{k} = n2^{n-1}$ counts committees with a chair: choose the chair (n), then any subset of the rest (2^{n-1}). Engines and stories are two proofs of one fact — quote whichever the question rewards.

KEY RESULT

Engine 3: products of coefficients are one coefficient

Vandermonde: $\sum_k \binom{r}{k}\binom{s}{m-k} = \binom{r+s}{m}$ — the x^m coefficient of $(1+x)^r(1+x)^s$, or committees from two groups. Specialisations: $\sum\binom{n}{k}^2 = \binom{2n}{n}$ (at $n=10$: 184756; at $n=6$ via $\binom{6}{k}\binom{6}{6-k}$: $\binom{12}{6} = 924$); $\sum\binom{n}{k}\binom{n}{k+1} = \binom{2n}{n+1}$; alternating squares $\sum(-1)^k\binom{n}{k}^2$ is the x^n coefficient of $(1-x^2)^n$: 0 for odd n, $(-1)^{n/2}\binom{n}{n/2}$ for even n. **Hockey-stick:** $\sum_{i=r}^{m}\binom{i}{r} = \binom{m+1}{r+1}$ (telescope Pascal), e.g. $\binom{4}{4}+\binom{5}{4}+\binom{6}{4}+\binom{7}{4} = \binom{8}{5} = 56$.

COMMON TRAP

Common traps — identities

- Differentiating then evaluating at the wrong point, or dropping the outer x in $xf'(x)$ (a factor-x error away from $x=1$: $\sum k\binom{n}{k}2^k = 2n\cdot 3^{n-1}$).
- Reaching for memorised identities when the engine derives them in one line — engines survive twists, lists do not.
- Vandermonde index mismatch: the second factor must be $\binom{s}{m-k}$ with the same k; check the target slot first.

LESSON 04 / 04

The ratio walk to the greatest term

KEY RESULT

Greatest term of $(a+bx)^n$ at a given x

For terms $t_k = \binom{n}{k}a^{n-k}(bx)^k$, walk the ratio $\frac{t_{k+1}}{t_k} = \frac{n-k}{k+1}\cdot\left|\frac{bx}{a}\right|$ until it crosses 1. Greatest term of $(2+3x)^8$ at $x=1$: ratio $\frac{8-k}{k+1}\cdot\frac{3}{2} \geq 1$ iff $k \leq 4.4$, so terms rise through t_5 and fall after: $T_6 = \binom{8}{5}2^3 3^5 = 108864$ (neighbours 90720, 81648 ✓). Use absolute values: "numerically greatest" means largest $|t_k|$.

KEY RESULT

Truncation: positivity turns expansions into inequalities

For $x \geq 0$ every term of $(1+x)^n$ is non-negative, so any partial sum is a lower bound. Keeping $k \leq 1$: **Bernoulli** $(1+x)^n \geq 1+nx$; keeping $k \leq 2$ sharpens it by $\frac{n(n-1)}{2}x^2$. Bernoulli in fact holds for all $x > -1$ (induction proof — the truncation argument alone covers only $x \geq 0$ since odd terms flip sign).

The same term comparison traps $\left(1+\frac{1}{n}\right)^n$: its k-th term is $\frac{1}{k!}\prod\left(1-\frac{j}{n}\right) \leq \frac{1}{k!} \leq \frac{1}{2^{k-1}}$, giving $2 \leq \left(1+\frac{1}{n}\right)^n \leq 3$ — e appears as the pinched limit.

KEY RESULT

Remainders and last digits: expand around the modulus

To compute A^n modulo m: write A as (multiple of m) $\pm$ small residue and expand — all terms carrying the multiple vanish mod m except the last few. Last digit of $7^{20} = 49^{10} = (50-1)^{10}$: every term except $(-1)^{10}$ carries a factor 50 (the $j=1$ term is $-500 \equiv 0 \bmod 100$), so last digit 1. Similarly $6^{999}+1 \equiv 1+1 = 2 \pmod 5$ since $6 \equiv 1 \pmod 5$.

KEY RESULT

Conjugate surds: integer parts without decimals

$(1+\sqrt{2})^n + (1-\sqrt{2})^n = 2\sum_j \binom{n}{2j} 2^j$ is an integer (odd $\sqrt{2}$ powers cancel by the binomial theorem itself), and $|1-\sqrt{2}| < 1$ leaves only dust. For odd n the dust is negative, so the integer part of $(1+\sqrt{2})^n$ is that integer: $(1+\sqrt{2})^3$ has conjugate sum $2(1+6) = 14$, and integer part 14 (even, for every odd exponent). For even n the dust is positive and the integer part is the conjugate sum minus 1 (odd).

COMMON TRAP

Common traps — size questions

- Truncation claimed for negative x without the induction form of Bernoulli.
- Ratio walk without absolute values when terms alternate in sign.
- Modular truncation treated as approximation: it is exact in residue classes — but every dropped term must genuinely carry the modulus factor.